



TiCorporate Portfolio

Exsanguination

Aapo Vanhainen

Portfolio

FF Ticorporate DemoLab 2, Jere Lamberg

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Degree Programme: Business Information Technology

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1 Project Background, Vision and Goals

1.1 Project background

Project is a game released to *Steam*, created by a small group of students, studying their third year in *Jyväskylä University of Applied Sciences* in the field of Game Development. The project is part of the course called *FF Ticorporate Demo Lab* consisting of 25 credits, that is 675 hours of total work. The project includes two courses related to business with 5 credits in total called *Business Basics* and *Business Communication* that are done concurrently with the game project.

1.2 Vision

At the early days, during prototyping and pre-production the vision for the game *Exsanguination* was as follows:

You are a vampire hunter, who freshly got turned into a vampire. Being a vampire isn't going to stop you from killing other vampires, merely it's an improvement over your previous situation. Ex-

plore your new abilities as you take down hordes of vampires, hellbent on killing you, an abomination. The world is dark and gloomy, and you must grapple with your new reality. Is there any escape or are you forever doomed to be the monster you swore to destroy?

As the productions developed into the final version, the vision was according to *Steam's About this game* section:

Exsanguination is a fast-paced, first-person melee combat game where you play as a former vampire hunter now turned into a vampire. Move quickly and masterfully using your grappling hook and challenge yourself with a speed-run timer for an added layer of excitement.

Will you survive the night or become a victim of the very creatures you once hunted?

1.3 Goals

Overall goals for the game project were to create a short and sweet game with interesting mechanics, good combat, and an epic boss fight. Emphasis on audio effects, visual effects, music, and lore were important to keep the game intact, giving the project much-needed game feel for a fast-paced hack and slash game loop.

Other goals regarding the whole project were: learning teamwork; working in a professional environment with team members; having individual responsibilities and areas of expertise, such as programming or design; dividing project into pre-production, production, post-production, and release, with each phase having its specific tasks in mind; learning to deal with surprises, such as mentoring a junior programmer and creating a whole *Steam* page in a very limited time.

1.4 Project Organizations

Project organizer was *Jyväskylä University of Applied Sciences*, granting us, *Never Late Productions*, the opportunity to work on this project and release it to *Steam*.

The people who took part in the project are as follows:

Aapo Vanhainen	Producer, Design, Audio, Programmer, Implementation
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Emilia Hietikko	Programmer, Tech Artist, VFX, Implementation, Design, Scrum Master
Charles Silvennoinen	Product Owner, Visual Design, Level Design & Implementation
Kimmo Riipinen	Lead Programmer, QA Lead, Audio
Vili Jalonen	Business, Narrative, Implementation and a bit of testing
Jesse Kauhanen	Junior Coder, QA-testing
Sanni Ouli	Level & Menu Music
Severi Boesen	Boss Music

1.5 Steering Group

In the steering group were the lecturers and interns from *Jyväskylä University of Applied Sciences* responsible for the whole course: Risto Koskenkorva, responsible for technology and implementation; Jere Lamberg, responsible for running the course and scrum; Aaron Wirenius, responsible for producing, publishing, and design; Mika Karhulahti, responsible for graphics and art; Kalle Raijoniemi, responsible for audio and QA/testing; Ilari Miikkulainen, responsible for narrative; Antony Smal, responsible for business.

2 Introduction

This document discusses experience of creating the game *Exsanguination* from my, Aapo Vanhainen's, point of view as being the designer, audio, publishing & producing responsible for the project. As the production commenced, my roles kept spreading to different areas, mainly implementation and programming.

For the project and course, itself we had multiple assignments, including hundreds of pages of PowerPoint slides, hundreds of combined hours within the team to combine into 2000+ man hours we spent on the project itself. In this document I'll encompass all the significant and noteworthy personal accomplishments and learning lessons.

Portfolio will go through the topics in following order: design, audio, producing & publishing, programming & implementation and miscellaneous other tasks.

3 Portfolio

3.1 Design

Design is one of my major areas of interest in game development, something I wanted to fully focus on during this project alongside audio. As a game designer I have my interests in game mechanics design, gameplay design, level design, enemy and combat design.

During our production of the game *Exsanguination* our team had three distinct designers: me, Emilia and Charles. We all helped each other and worked on every decision, but our main focuses were as follows: mine in gameplay, combat and enemies, Emilia's in enemy, narrative implementation and visual effects and Charles' in level design and graphics/art design. We all worked on game mechanics together and as with every aspect, we took other team members outside of design into the discussion. Our meetings related to the game design and the document, as well as MVP, were mostly in the early stages of the production, meaning in pre-production and first few weeks of production. After that the decisions were discussed on the fly either in the office or on Monday mornings or Friday evenings if we reserved time to sit down together and talk.

Game design document, GDD, was major part of the TiCorporate course and whilst we felt it was very useful alongside MVP in the beginning, as soon as the production started so much changed and we had to constantly update the GDD it was quite overwhelming. I personally find GDD very useful even when things change, wish I would've had more time working on the actual document and not leaving it half finished midway through. All of us were busy on working on our own, and me whose job it was to update it, had to work on much more and different aspects of the game (chapters 3.6 and 3.7) that I simply did not have time.

3.2 Audio

Audio was my second major areas of game development where I worked with two talented music producers, two team members, one member of the other project and our teacher during this project. I worked on both sound effects with Kimmo and music with Charles and the music producers. Most of the sound effects were recorded in a recording studio with Kimmo but some were taken online due to time constraints or being too difficult to self-record. For boss chants we asked our classmate Milla if she wanted to help, and she agreed happily.

3.2.1 Music

We got lucky and got two music producers to help create the music for our game, Severi and Sanni. We had weekly meetings with Charles and them, discussed what we wanted for the game and what they wanted to produce. Severi wanted to focus on the boss music and Sanni on the level music. Initial idea of the music in the level was as follows: player is not in combat; play slow paced but anticipating music, player is in combat; play fast paced and Doom-like music. Unfortunately, we were unable to get this experience and were left off with the anticipation music only but hearing feedback from players they seem to like it very much.

Boss music arrived on time and with intro and 4 different intensities for different phases of the boss. Implementing them took a lot of time but all of them are in the game and they sound very good.

Level music on the other hand was handled very poorly. Communication with Sanni ended abruptly couple weeks before the intended return date, we didn't hear from her until three days *after* the deadline where it was almost too late to implement the music for one of the project's biggest testing sessions, Day Zero. I had to scramble through .mp3 file in Audacity to cut it to perfect loop, create an intro and then implement into the game. It was very stressful and difficult to work only on one audio file without having it separated into different files as we discussed. Also the latter part of the music was something we (and later heard she too) were not satisfied and we had to work with the intro part of the song. If we could've discussed with her about the music beforehand we surely could have made improvements. The second song she created that was used in main menu and death screen was good and we implemented it as is.

For all of the music I spent a lot of time in Visual studio making the musicmanager script (more on this in Chapter 3.6.3) and in Unity with volume masters, making sure the different sliders work in the settings and to find the good volume for the music in game.

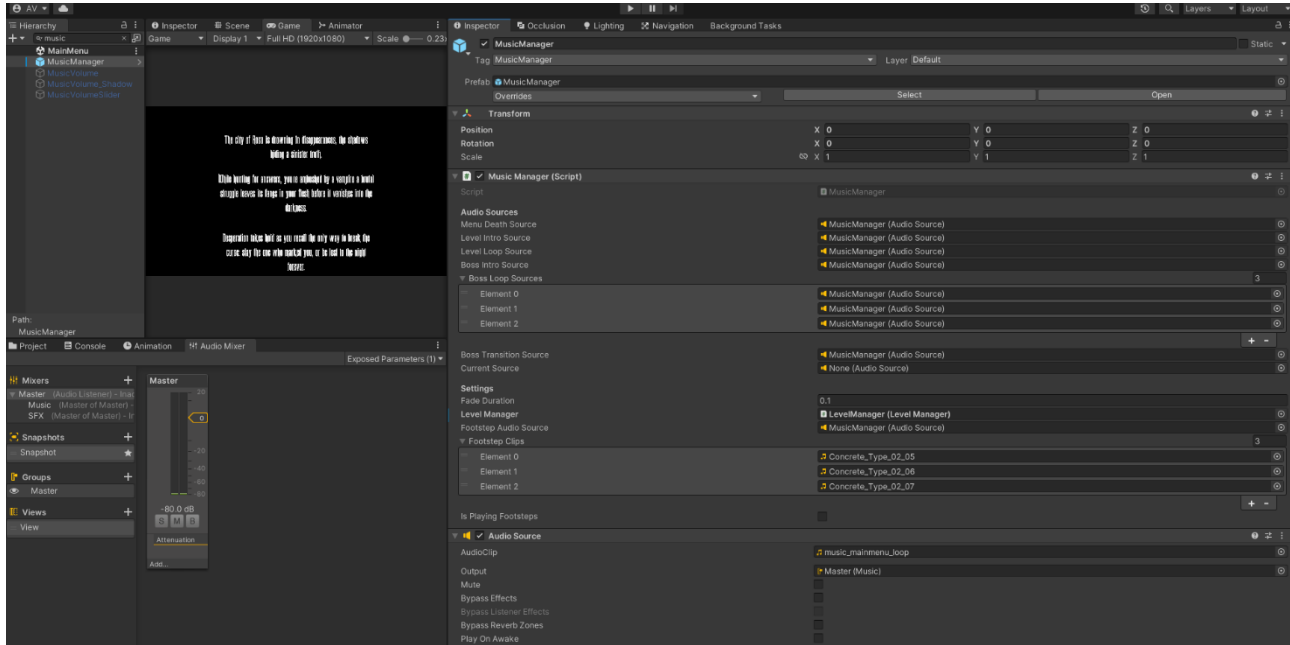


Figure 1: MusicManager and Audio Mixer in Unity.

3.2.2 Sound effects

Sound effects, or SFXs, were done with the help of our lead programmer Kimmo. We started working on Excel sheet on what sounds we need for our game, what we should record ourselves and what sounds should we get online, I also started working on Audio Design Document, ADD, for both music and sound effects. The ADD ended up focusing on music and the excel for sound effects, although such things as naming convention when importing them in-game are found from ADD.

We worked in a recording studio twice and using tools such as actual swords, pliers, shoes and our mouths we created sound effects for the game. Recordings were saved in Audacity to be later modified. Custom sounds were recorded for almost every sound in the game including sword

swings, enemy alert sounds, player slide, player jump, grappling hook hit and all the boss chants during special attacks. Other sounds were gathered from online libraries.

Working on sound effects meant more than just recording. They required editing such as removing background noise, normalizing, reverbing and more. Also, some of the sounds needed one or two more variants so they don't sound weird when played multiple times in a row such as sword swing. This included editing sounds from online libraries to make variants of the same sound. The work was done in Audacity since it provided all of the necessary tools for it.



sfx_Energy_Bat_alert_var1.ogg

Audio 1: Self recorded screech when bat notices the player and starts attacking.



sfx_Combat_StarterSword_Swing_3_var2.ogg

Audio 2: Variant 2 of the third swing of the sword-combo, recorded with training sword.



sfx_Player_Jump.wav

Audio 3: Player jump sound, recorded with flinging t-shirt upwards.



sfx_Boss_SpecialAttack_Spike.ogg

Audio 4: Boss chant for spike special attack, edited from original recorded audio.



20241022_151528.mp4

Video 1: Recording sword equip sound in recording studio.



Figure 2: Me and Kimmo recording sword equip sounds.

3.3 Producing & Publishing

3.4 Producing

Being a producer meant I was responsible for making sure everyone had tasks to do, and that the tasks they were doing were relevant to the current state of the project. Meaning that the tasks team members were doing were not outdated or irrelevant. This was done discussing in the office or via Discord messages if either one was not in the office. Discord had multiple channels for information for example for every speciality such as design, graphics, narrative, level design, programming etc. and channels like schedule, announcements, idea dump and memes. Discord was also used to create weekly or daily tasks for everyone. The tasks were more open-ended as the project began but as the game production progressed, the tasks got more and more precise. All of the decisions regarding tasks and project work were discussed with everyone, or at least with the affected members.

Another part of producing was making sure everyone was feeling okay and not overwhelmed by their tasks. I personally think I fit for this role very well and was thriving in it. Although checking up on myself was sometimes neglected.

Also, as a producer I had to make the final say when something was either left hanging or had unbalanced opinions whether on how to do it. This meant that in cases where I was on the opposite end and the majority were on the other, I would trust my team and go with the vote of the majority, and I'm glad I did. Also, when the decision of design, mechanic or such was more of a 50-50, I tried to find the best solution by having best of both worlds, or just making a decision and choosing one of the other. For example, I personally wanted to have at least two swords in the game but as the production went on, the time constraint was so tight that after discussing with other designers we came to an agreement to leave the game as is, with just the one sword.

Aapo Vanhainen 16/09/2024 14:36
🔥 Sprint week 1: 16 - 20.9🔥

@Kimmoke
Tuesday:
💻 Create new Unity project in Ticorporate repo
(<https://github.com/jamktiko>) 💻
📄 Write tech guidelines 📄
🔧 Add prototype stuff into new project 🔧
🚶 Create Rigidbody based movement with current skills: Slide and Dash. (Continue on Wed-Thursday) 🚶

Wednesday:
🎵 Audio meeting 11:00 🎵
🎵 With **@Aapo Vanhainen** SFX Design 🎵
🚶 Continue with Rigidbody based movement with current skills: Slide and Dash. 🚶

Thursday:
🚶 Continue with Rigidbody based movement with current skills: Slide and Dash. 🚶

@icxrus / Emilia
Tuesday:
👤 Finish managers (Save system etc.) 👤

Wednesday:
😓 WFH 😓
🔥 Narrative: Character Design 🔥

Thursday:
❓ VFX Design ?
👤 Enemy design with **@Aapo Vanhainen** 👤

@Aapo Vanhainen
Tuesday:
💻 Gameplay Design 💻

Wednesday:
🎵 Audio meeting 11:00 🎵
🎵 With **@Kimmoke** SFX Design 🎵
📄 Music Design 📄

Thursday:
📄 GDD writing and fixing 📄
👤 Enemy design with **@icxrus / Emilia** 👤

@Vilonen
Tuesday:
📄 Writing the story 📄

Wednesday:
👤 Boss backstory 👤
📄 Create main storyline 📄

Thursday:
📖 Lore books 📖

@Charles / fl0w
Tuesday:
👤 Concept art 👤

Wednesday:
🎵 Audio meeting 11:00 🎵
👤 Color palette 👤

Thursday:
📖 Art Bible 📖
📄 Level Design 📄

Figure 3: First sprint week schedule with very open-ended task lists.

Charles:

Make red torches something else color! Can not be seen clearly! Think contrasts!

add info lights (currently red) to tutorial and way more in the level (first staircase, gp point to the next platform, in the circle thing to show u go here up there)

Add boxes/wall smth so u need to slide into gp room

Kimmo:

Find better solution to gp slide, 0.1 drag does not work (player still slides over the edge in gph room)

Boss dagger attack never hits

Vents clearer?

Emilia:

Settings work with other scenes or we scrap them from mainmenu

Player prefs not working?

Sword hit 2&3 vfx smaller / not take so much screen space

Jesse:

Test

Test like normal player first, no dash fly away things just normally try to find Ur way forward

Vili:

Write 1-2 "slide shows" of text (few sentences) of why we r here, what we are here to do. Its shown before game starts after main menu

@everyone (edited)



Aapo Vanhainen 02/12/2024 10:44

@Aapo Vanhainen

Tutorial - stop advancing until enemy is killed (in exit script check if the enemy is disabled then can continue), start ienumerator for scene change after (5?) seconds, make screen go black, add door open + steps sound and after those go to Level

Remove traps from tutorial lol

Similarly when going staircase to mechtestroom add same effects but now with rb is kinematic, input disabled, teleport player -> rb nonkine and input enbled

Going to boss room add slower walking audio and bigger door open sound for cutscene (also fix the font issue on cutscene!)

Make dontdestroyonload music that works when resetting, with crossfade

Figure 4: Task lists when nearing the end of production.

3.5 Publishing

I was responsible for publishing of our product *Exsanguination*. This involved most notably creating the Steam page and working in Steamworks. This meant capturing the screenshots in game, creating capsules, making the trailer, writing the about the game, mature content description, filling all the necessary information such as system requirements, tags and more. Also, I was the responsible for adding new builds to Steam. Unfortunately, due to the nature of our course and instructions, I was given 1 day time to create the Steam page, and since we had planned to do it a bit later, the page creation was not even started at this point. Personally, I'm very disappointed and reckon the quality of the Steam page is mediocre at best.

3.5.1 Steam page

Link to the Steam page: <https://store.steampowered.com/app/3286090/Exsanguination/>.

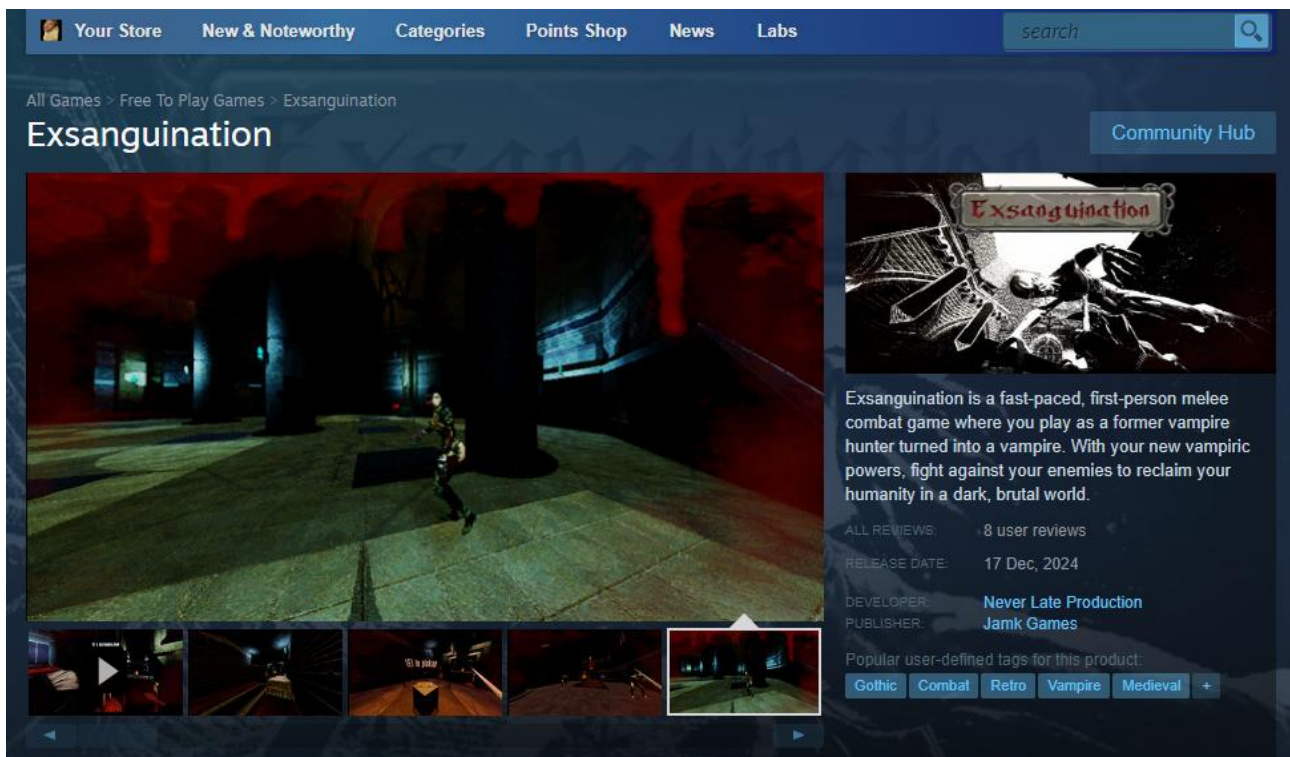


Figure 5: Steam page with screenshots and short description.

ABOUT THIS GAME

Exsanguination is a fast-paced, first-person melee combat game where you play as a former vampire hunter now turned into a vampire. Move quickly and masterfully using your grappling hook, and challenge yourself with a speed-run timer for an added layer of excitement.

Will you survive the night or become a victim of the very creatures you once hunted?

Features

Use your newfound vampiric powers to dash, slide, and engage in brutal melee combat against hordes of ghouls and bats. Block enemy attacks, reflect projectiles, and strategically use your wooden stake to heal yourself and finish off foes. Explore a dark, treacherous world filled with traps, secrets, and hidden challenges, including an epic boss battle with the mysterious figure who turned you into a vampire.

Credits

- **Charles Silvennoinen** - Product Owner, Environment and Level Design, Art Direction
- **Aapo Vanhainen** - Producing, Publishing, Design, Audio, Programming
- **Kimmo Riipinen** - Lead Programmer, Audio, QA Lead
- **Emilia Hietikko** - Programming, Visual Effects, Design, Scrum Master, Narrative
- **Vili Jalonen** - Narrative, Business
- **Jesse Kauhanen** - Junior Programmer, Testing

Special thanks to Sanni Ouli for producing the level music and Severi Boesen for producing the boss fight music for this game.

*This game is a student project created in a period of 9 weeks.

MATURE CONTENT DESCRIPTION

The developers describe the content like this:

This game features intense violence, including graphic depictions of enemies exploding with blood and gore, leaving visceral stains on the environment. The sound design emphasizes the brutal nature of combat, with effects that suggest slicing through flesh and violent dismemberment. Some enemy character designs include revealing outfits, with partial nudity.

Single-player
 Steam is learning about this game

Languages:

Interface
 Full Audio
 Subtitles

 English

STEAM DECK COMPATIBILITY

Unknown
 [Learn more](#)

TITLE: Exsanguination

GENRE: Action, Adventure, Indie, Free To Play

DEVELOPER: Never Late Production

PUBLISHER: Jamk Games

RELEASE DATE: 17 Dec, 2024

[View update history](#)
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Figure 6: Steam page information

Trailer was, as everything else, done in a hurry with OBS to do the recording and Movie Maker to edit it. It contains out-dated gameplay and old menu screen. As the project started to end, I had a decision to make: make better steam page and trailer or fix in game bugs; I chose fixing bugs. Trailer for the game can be found from Appendices.

3.6 Programming & Implementation

Even though programming was not part of my plans for TiCorporate and implementation being only a support role, I started working on them profoundly. As we noticed very early on that we don't have time to finish everything we wished for with the two programmers I stepped in. I personally feel like I'm not as good as programmer as the others, but I managed to work in a vacuum to create mostly everything, until I didn't...

Major features for programming and implementing I worked on *Exsanguination* were creating the player character, two distinguish enemies, finding and giving level assets to level designer, various level additions such as reset points, smoke, creating lightning, changing grappling hook notification logic, creating enemy ai pathfinding and more.

3.6.1 Player

For the player, I handled finding and editing the model, creating combat logic including blocking, reflecting projectiles, dealing damage, and combo mechanics, implemented movement and sword combat animations with directional movement, sliding, and a three-hit combo, sourced and integrated the sword model, and fixed various issues across multiple scripts.

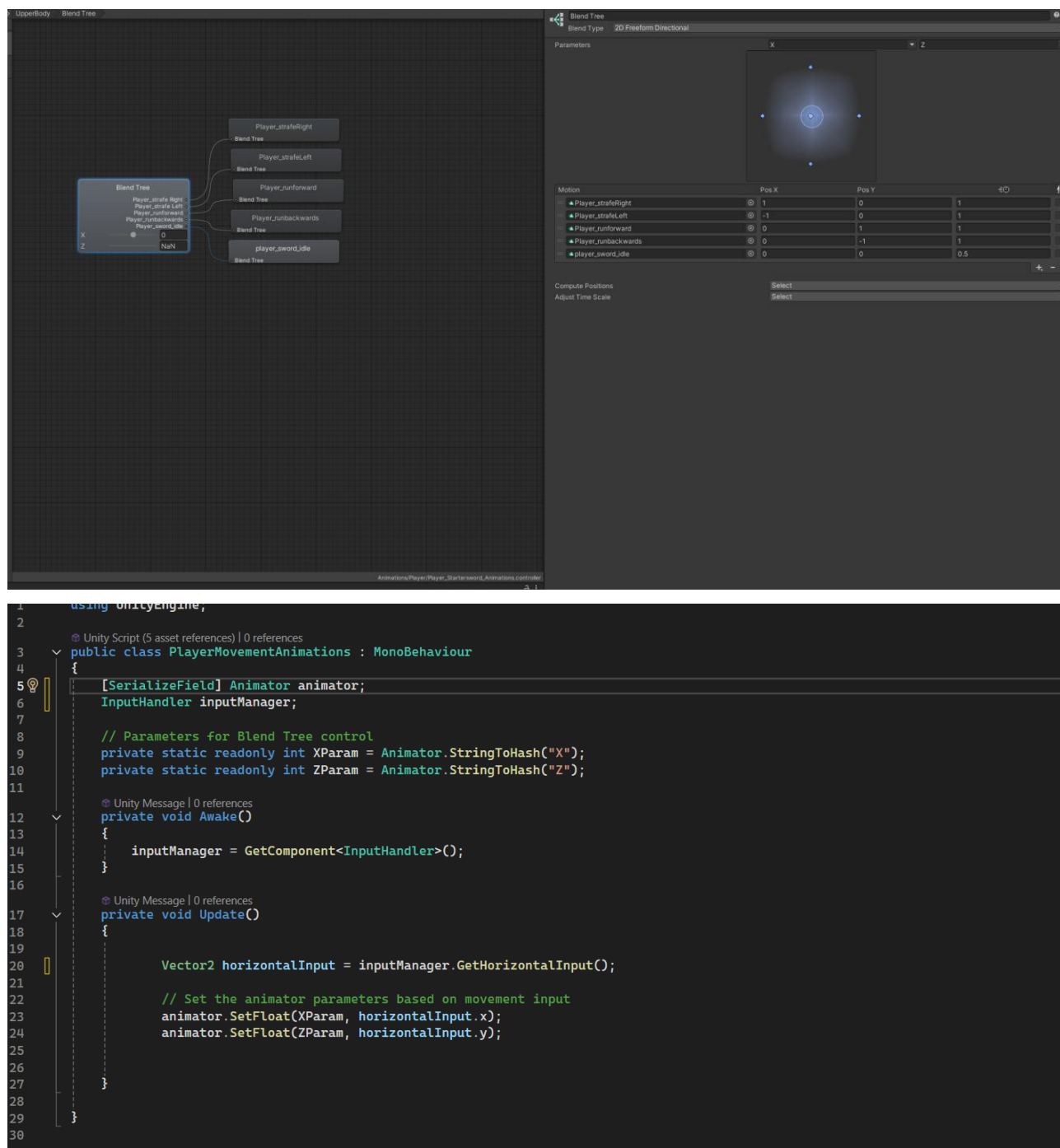


Figure 7: Blend tree for 4-way movement animations and script to handle it.

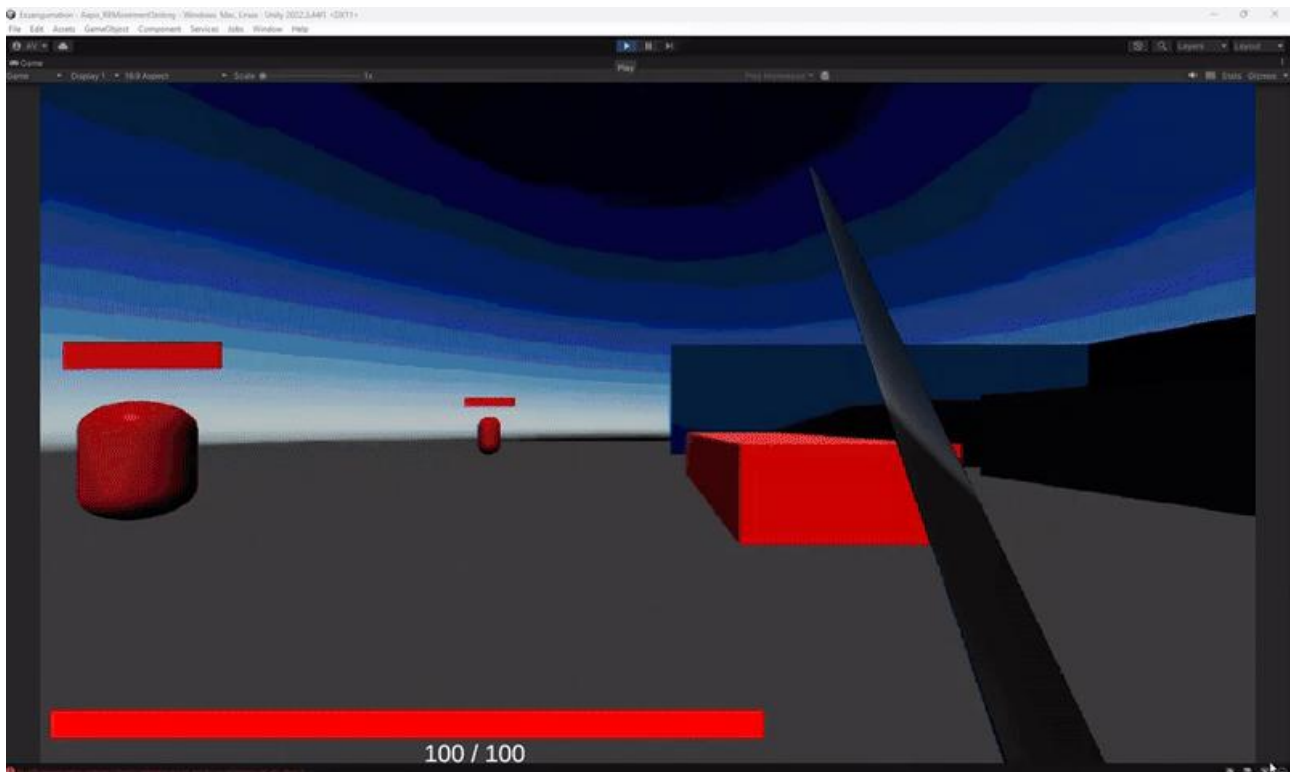


Figure 8: Player 4-way movement animation in early testings.

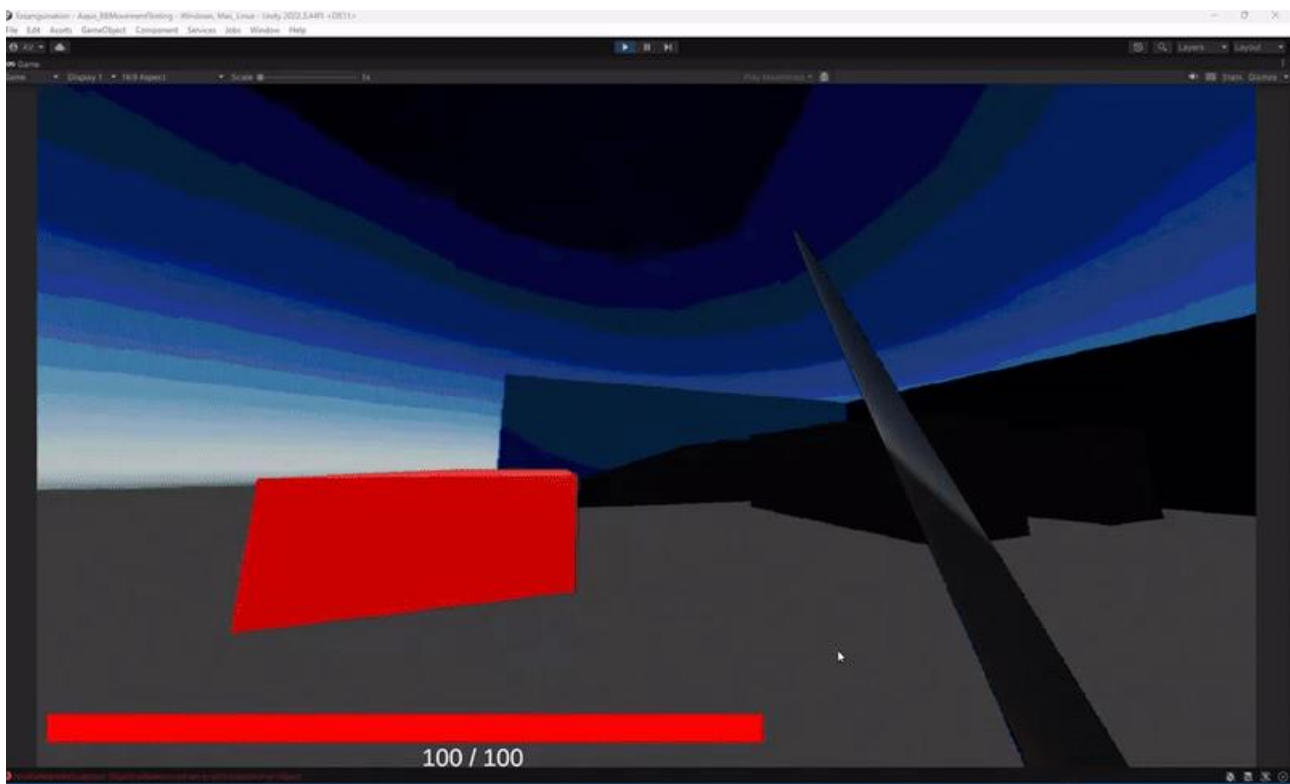


Figure 9: Player combosystem in early testings.

3.6.2 Enemies

I created two different enemies: ghouls and bats. These I created all by myself and all of their features, logic and animations etc. The 3D models were from an asset and personally I'm pleased that they seem to fit in the level and graphics very well even everything including level, enemies, player, weapons etc. were from different assets banks.

For enemies I created all the logic for being idle, chasing player, attacking, reaching the gap between them and player with a pounce (which was deprecated at the end stages of production due to having no time smoothing out the animations, it ended up looking a bit weird), being stunned, dying and damaging the player.

I created the animations with the help of Mixamo: idle, chase (run), attack1, attack2, stunned, jump (if needed to reach higher points, this logic was deprecated also due to production time).

The process was extensive with a lot of testing and asking for feedback, finding and fixing bugs and whenever something new was implemented it had to be tested and fixed accordingly.

1 reference

```

void AvoidOtherEnemies()
{
    // Only run this check every 0.2 seconds to reduce overhead
    if (Time.time < nextSeparationCheckTime) return;
    nextSeparationCheckTime = Time.time + 0.2f;

    // Reset the accumulated separation force
    separationForce = Vector3.zero;

    Collider[] hitColliders = Physics.OverlapSphere(transform.position, separationDistance, enemyLayer);

    foreach (var hitCollider in hitColliders)
    {
        if (hitCollider.transform != transform) // Ignore self
        {
            // Accumulate direction to avoid other enemies
            separationForce += (transform.position - hitCollider.transform.position).normalized;
        }
    }

    // Apply the accumulated separation force scaled by moveSpeed and delta time
    rb.AddForce(separationForce * moveSpeed * Time.fixedDeltaTime, ForceMode.VelocityChange);
}

```

1 reference

```

void FollowPlayer()
{
    float distance = Vector3.Distance(transform.position, player.position);
    Vector3 direction = (player.position - transform.position).normalized;

    if (!enemyStates.isStunned && !enemyAnimator.GetBool("isAttacking") && distance > attackRange && navMeshAgent.enabled && isGrounded && !isPouncing)
    {
        RotateTowardsPlayer(direction); // Always rotate towards the player
        navMeshAgent.SetDestination(player.position); // Use NavMesh to move towards player
    }

    // Check if the player is within attack or pounce range
    if (distance <= attackRange)
    {
        if (CanAttack() && !enemyStates.isStunned)
        {
            SnapRotationTowardsPlayer(direction);
            Attack(); // Trigger the attack
        }
    }

    if (pouncingEnemy && distance <= pounceRangeMax && distance >= pounceRangeMin && CanPounce() && !enemyStates.isStunned && navMeshAgent.enabled && !enemyAnimator.GetBool("isAttacking") && !isStuckOnStake)
    {
        SnapRotationTowardsPlayer(direction);
        if (isGrounded)
        {
            Pounce();
            lastPounceTime = Time.time;
        }
    }
}

if (distance <= stopSeparationDistance)
{
    separationDistance = 0;
}
else
{
    separationDistance = storedSeparationDistance;
}
}

```

```

1 reference
void Attack()
{
    // Check if the enemy is already attacking
    if (enemyAnimator.GetBool("isAttacking"))
        return; // If attacking, do nothing (don't reset or start another attack)
    lastAttackTime = Time.time; // Record the last time an attack was initiated
    enemyAnimator.SetBool("isAttacking", true); // Start attack animation
    StartCoroutine(ResetAttackAfterCooldown()); // Reset attack state after the animation
}

1 reference
IEnumerator ResetAttackAfterCooldown()
{
    yield return new WaitForSeconds(attackCooldown); // Wait for attackCooldown duration
    enemyAnimator.SetBool("isAttacking", false); // Reset attacking state
}

1 reference
void Pounce()
{
    pounceDirection = (player.position - transform.position).normalized;
    pounceDirection.y = pounceForceUp;
    enemyAnimator.SetTrigger("pounce");
    isPouncing = true;
}

1 reference
bool CanPounce()
{
    return Time.time >= lastPounceTime + pounceCooldown;
}

1 reference
bool CanAttack()
{
    return Time.time >= lastAttackTime + attackCooldown && !enemyAnimator.GetBool("isAttacking");
}

1 reference
void RotateTowardsPlayer(Vector3 direction)
{
    // Calculate the rotation needed to face the player
    Quaternion lookRotation = Quaternion.LookRotation(new Vector3(direction.x, 0, direction.z));

    // Smooth rotation
    transform.rotation = Quaternion.Slerp(transform.rotation, lookRotation, Time.fixedDeltaTime * 5f);
}

```

Figure 10: EnemyAI methods to avoid clumping, chasing and attacking.

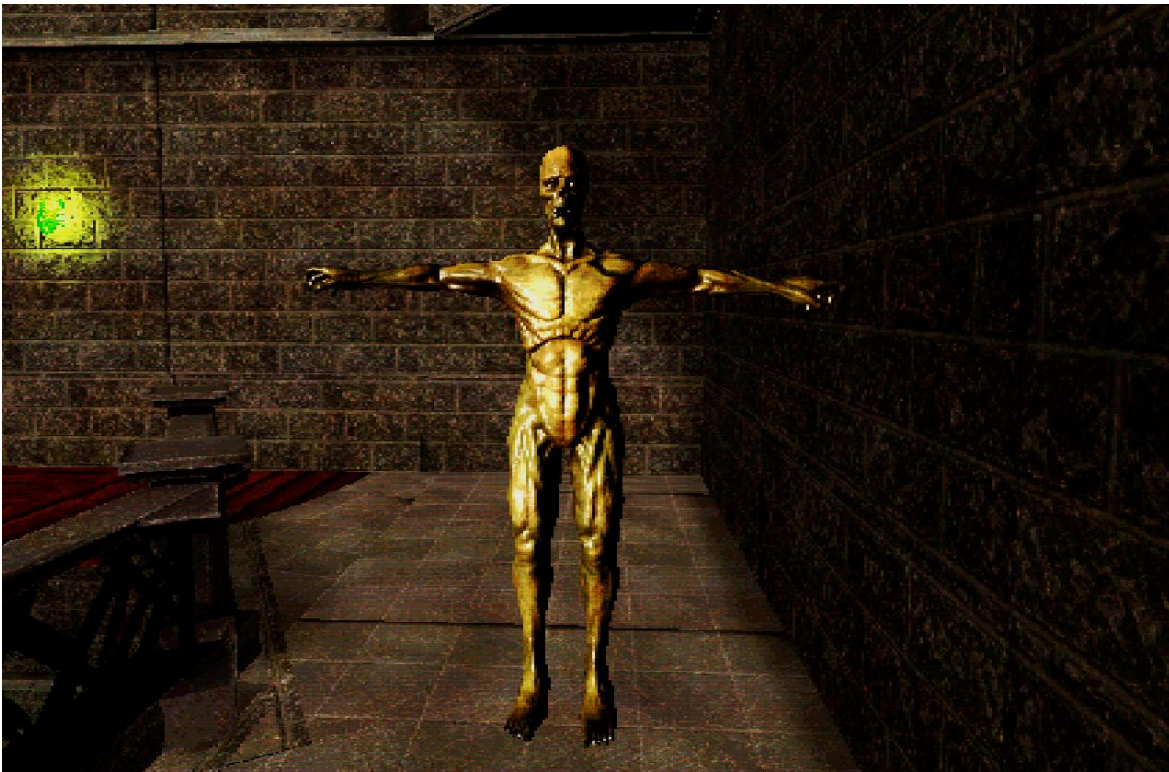


Figure 11: Two enemies, ghoul and bat.

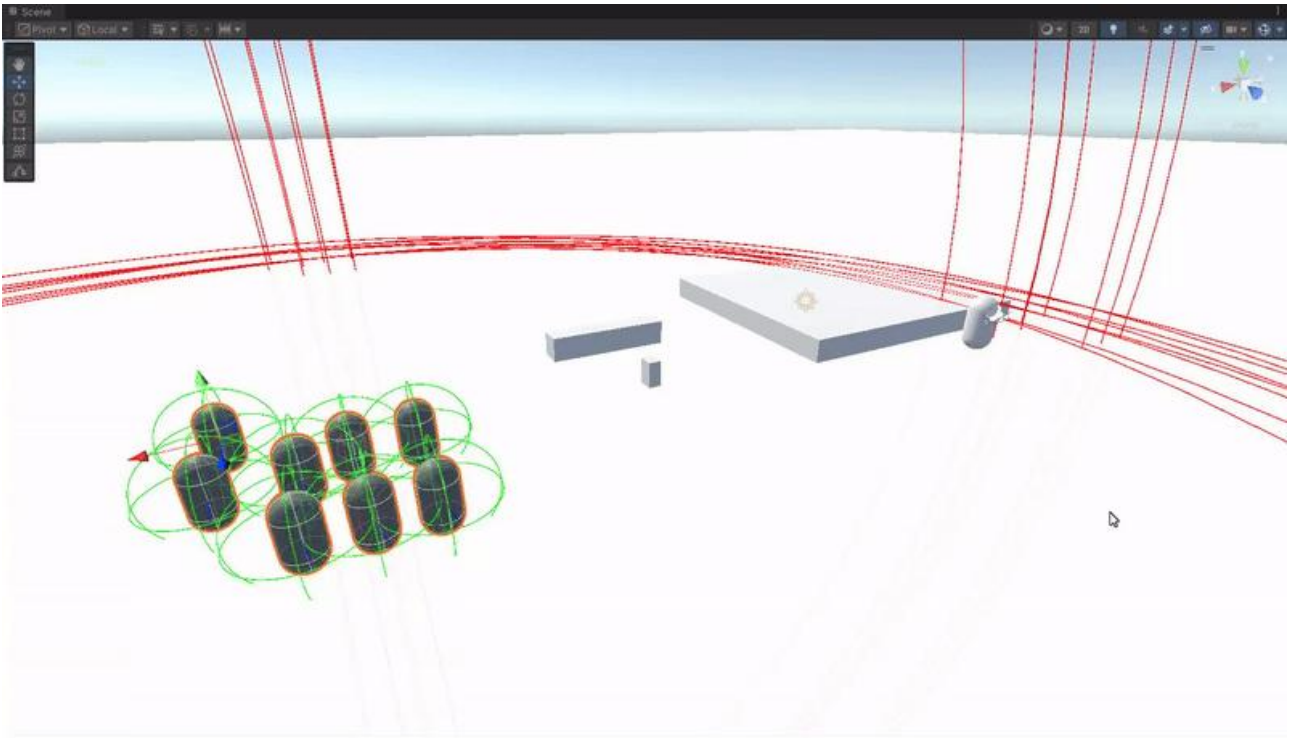


Figure 12: Enemies avoid clumping when chasing in early testings.

3.6.3 Miscellaneous

Other miscellaneous programming and implementations included the following: adding health bar in to the crosshair, various daily level fixes (the levels needed fixing all the time every day), enemy navmesh AI pathing in the actual levels, player reset boundaries, smoke effect for pits and enemy dodging phase, creating new system for aiming the grappling hook lights, creating logic for picking up the key and replacing it multiple times since testers did not find it or were unable to do so, changing enemy positions and amounts in level and more.

More sizeable task was re-creating the lightning, baking it and re-doing it multiple times (also learning how to do lightning in Unity as I had only little experience beforehand). Another time consuming task was creating the cutscenes (and rewriting them completely), making sure they work each time but also when player dies in the boss fight it skips the cutscene and starts the fight immediately. Luckily the logic for cutscene was written by our junior programmer and I only needed few small fixes to that to make it work. Lastly creating the music manager. That took a lot of time and fixing since there were different scenarios when the music should start and stop, where the loop comes in and so on.


```

using UnityEngine;

Unity Script (1 asset reference) | 3 references
public class CutsceneSkip : MonoBehaviour
{
    public static CutsceneSkip Instance;
    public bool hasSeenCutscene = false;

    Unity Message | 0 references
    private void Awake()
    {
        if (Instance == null)
        {
            Instance = this;
            DontDestroyOnLoad(gameObject); // Keeps this object across scene loads
        }
        else
        {
            Destroy(gameObject); // Destroy duplicate instance if one already exists
        }
    }

    1 reference
    public void MarkCutsceneAsSeen()
    {
        hasSeenCutscene = true;
    }
}

Unity Message | 0 references
private void Start()
{
    if (!cutsceneSkip.hasSeenCutscene)
        StartCoroutine(TransitionInBoss());
}

```

Figure 13: Simple logic to cutscene skip: have a gameobject that is not destroyed when reloading the scene in boss fight and in cutscene script have check if the player has seen the cutscene then dont run the cutscene.

```
// Play Level music: Crossfade from current track to level intro, then immediately to loop
2 references
public void PlayLevelMusic()
{
    CrossfadeToSource(levelIntroSource, 2);
    currentSource = levelIntroSource;
    StartCoroutine(HandleLevelIntroAndLoop());
    isBossMusicActive = false;
}
```

```
1 reference
private IEnumerator HandleLevelIntroAndLoop()
{
    if (!levelIntroSource.isPlaying)
    {
        levelIntroSource.Play();
    }
    // Wait for intro to finish before starting the loop music
    yield return new WaitForSecondsRealtime(levelIntroSource.clip.length);
    levelLoopSource.Play();
    currentSource = levelLoopSource;
}
```

```
// Play boss music: Crossfade to intro, then loop music with variations
1 reference
public void PlayBossMusic()
{
    if (isBossMusicActive) return;

    isBossMusicActive = true;
    StartCoroutine(HandleBossIntroAndLoop());
}
```

```
private void CrossfadeToSource(AudioSource newSource, float duration)
{
    if (currentSource != null)
    {
        // If there is a current source, start fading it out while fading in the new source
        StartCoroutine(FadeOutAndIn(currentSource, newSource, duration));
    }
    else
    {
        // If there is no current source (initial track), just play the new track
        newSource.Play();
        newSource.volume = 0.5f; // Set a default volume for the new track
        currentSource = newSource; // Update the current source reference
    }
}
```

```

1 reference
private IEnumerator FadeOutAndIn(AudioSource fadeOutSource, AudioSource fadeInSource, float duration)
{
    // Save the starting volume of the old track (fade-out)
    float startVolume = fadeOutSource.volume;
    float elapsedTime = 0f;

    // Ensure the new source starts playing, but with zero volume initially
    fadeInSource.volume = 0f;
    fadeInSource.Play();

    // Fade out the old track and fade in the new track simultaneously
    while (elapsedTime < duration)
    {
        float t = elapsedTime / duration;
        fadeOutSource.volume = Mathf.Lerp(startVolume, 0f, t); // Fade out the current track
        fadeInSource.volume = Mathf.Lerp(0f, 0.5f, t); // Fade in the new track
        elapsedTime += Time.unscaledDeltaTime;
        yield return null;
    }

    // After the fade duration, stop the old track and ensure volumes are correct
    fadeOutSource.Stop();
    fadeOutSource.volume = 0f;

    // Set the final volume for the new track
    fadeInSource.volume = 0.5f; // Set to your desired volume (can be adjusted)
    currentSource = fadeInSource; // Update the current source to the new track
}

```

Figure 14: Example methods from MusicManager script.

3.7 Miscellaneous tasks

I mentored a junior programmer midway through, creating tasks, coordinating cooperative tasks with team members like the lead programmer, reviewing test session notes, and planning future tests. I also actively participated in testing by acting as a tester, providing written feedback, and reviewing feedback.

3.8 End spaghetti

At nearing the end of the production, we still had so much work to do and so little time, I had to create not so clean or great code but something that at least works. This caused a bit of a mess since other team members would be unable to either read it or needed help in implementing their own work and I feel very sad about it. Although we managed to finish the project and implement our own things, it still caused some stress but it was a necessary evil. This caused some convoluted prefabs and scenarios where multiple people had to work on a same scene, waiting for one to finish the work first.

4 TiCorporate: self reflection

4.1 What did I know at the beginning

At the beginning, I had a clear understanding of my roles and responsibilities in the project. My main focus was on game design, where I worked on creating the Game Design Document, defining gameplay elements, and designing mechanics like movement, combat, and player feedback systems. I also handled programming tasks such as enemy AI and player combat logic, as well as implementing 3D models, animations, music, and sound effects.

I chose audio as a specialty because I wanted to explore my interest in music and sound design, working closely with others to implement and create music and effects for various gameplay elements. For project administration, I stepped into a producing role, focusing on project phasing, timelines, task management, and maintaining the team's workflow. Although it wasn't my primary area of expertise, I saw it as an opportunity to learn. I also chose this for my interest towards creating a Steam page, fixing it and learning how to publish.

I anticipated learning real-world applications for game development, teamwork, and avoiding feature creep while getting better as a designer and Unity developer. I also aimed to create a portfolio showcasing teamwork, a finished game project, music and audio work, and design documentation. This project felt like a chance to grow and prepare for a professional future in game development.

My actual skills were lacking in publishing and producing field, had some experience with audio and music, and few years of programming and fiddling around with Unity. I felt like my strongest skills were people skills, implementing in Unity and project leading.

4.2 What I know now

Game projects are tough. Good planning helps, but it's not enough—you need solid execution too. Scrumming and task management are super important to keep things moving. We learned to work

in a structured way, focusing on programming, implementing, and tackling tasks step by step. Cutting features early (killing babies) was crucial. We had to remove big features, trim smaller ones, reduce the number of weapons, enemies, and levels, and shrink the MVP as much as possible. Doing this early made the project manageable.

Teamwork is hard but worth it. When everyone's learning at the same time, it's harder to get stuff done, but it's also more forgiving since we're all figuring it out together. Communication is everything. Team meetings can work if they're well-organized, but if they're not, they're just a waste of time. I bet in the real world there are a ton of pointless meetings too.

Overall, making a game is a lot of work, and it's all about balancing ideas, skills, and practicality. Looking back, we could've adjusted our scope better to handle absences, but we pushed through, even if it meant stepping out of our comfort zones. It's clear now how important it is to adapt and make tough calls early on.

In Unity implementation I have better understanding and now know more aspects about aspects like audio mixer, how to implement audio well, creating different animation systems, lightning and occlusion, and more.

4.3 Goals

Most goals were achieved, even if it required extra hours and effort. The team atmosphere remained positive, and working in an office with set hours helped maintain focus. Dividing responsibilities allowed members to work in their areas of interest, and we gained valuable insights into the game development cycle, including Steam publishing. The game itself was well-received, achieving decent player engagement and positive feedback despite minor bugs.

However, the project faced challenges. As deadlines approached, the codebase became increasingly disorganized, leading to difficulties in debugging and scene management. Prefab usage caused frequent frustration due to fragile setups, and last-minute bugs created additional stress.

The unexpected addition of a junior programmer was a significant setback, requiring time for training and task creation, which felt unfair and poorly managed. Team absences also forced others to take on more work than planned. In hindsight, reducing the project scope could have alleviated some of these pressures, but key features were prioritized to maintain quality.

4.4 What did I learn outside of my learning plan for TC?

I learned how to mentor a junior, at least on some level. Even though it is not something I wish to do in the future or is my specialty, I am capable of it.

Discord was also something I also learned more and to use better during the production process. Editing the channel, creating pinned messages and more.

4.5 My future

As for the future, I feel like the TiCorporate course helped me a lot. One of the best advantages is to tell someone such as recruiter that I have created a game in Steam with a small team of students. I can tell what I did for the game, some challenges and how we overcame them. Even when the end product is not as great quality as some free games in Steam, it is still a project (done in very short period of time) that was done from start to finish.

4.5.1 Ideal future job

I don't know yet what my ideal future job is. I wish to do practical trainings and see how the real world of gaming industry is, what are the options and opportunities, where should I aim for. I really feel like my strongest suit is in design and people skills but have know-how in audio, programming and in implementing too.

An impossible ideal future job would be designer in either my own company with other members or in a big company like Valve working on Half Life 3 (which might be coming soon).

4.5.2 Did the course support my goals

I personally tailored my goals to fit the project as well as possible so they could be supported, and I feel like most of them did. Some mishaps and unnecessary or awful stuff was involved such as business basics, steam page incident, tons of assignments and presentations etc. I wish the course would've just focused on making the game, no unnecessary stuff on top of it, nobody reads the documents we did during the production, nobody will refer to them in the future and creating multiple 20+ page documents is insanity.

4.6 My strengths

My strengths are in people skills, being active part of the team, listening and communicating, eager to learn more and ask for help, interests in multiple fields and some know-how in many. I wouldn't say I'm jack of all trades but more than skills in one specific field. I'm also positive and try to think what's best for everyone equally and always learning more.

References

Pew Research Center. (2015). *Gaming and gamers*. <https://www.pewresearch.org/internet/2015/12/15/gaming-and-gamers/>

Appendix 1. Trailer

[https://video.fastly.steamstatic.com/store_trailers/257075919/movie480_vp9.webm?t=1733215911.](https://video.fastly.steamstatic.com/store_trailers/257075919/movie480_vp9.webm?t=1733215911)

Appendix 2. Sound effects Excel

Sound	Priority	Responsible	tmp ✓	Proper: R/I/E	Game-Ready?	Recorded	Low
Sword equip 1	Important	Aapo	✓	Record		✓	
Sword equip 2	Important	Aapo	✓	Record		✓	
Sword equip 3	Important	Aapo	✓	Record		✓	
Sword Swing_1	Important	Kimmo	✓	Record		✓	Medium
Sword Swing_3	Important	Kimmo	✓	Record		✓	Medium
Sword Swing_2	Important	Kimmo	✓	Record		✓	Medium
Weapon2 equip1	Important			Record		✓	
Weapon2 equip2	Important			Record		✓	
Weapon2 Swing1	Important			Record		✓	
Weapon2 Swing2	Important			Record		✓	
Weapon2 Swing3	Important			Record		✓	
Weapon2 equip3	Important			Record		✓	
Successful parry	Important	Kimmo	✓	Record		✓	
Silver bomb ignite	Important	Aapo	✓	Internet			
Silver bomb sizzle	Important	Aapo	✓	Internet			
Silver bomb explosion	Important	Kimmo	✓	Edit			Important
Grappling hook shoot	Medium	Kimmo	✓	Internet			
Grappling hook hit	Medium	Kimmo	✓		✓		
Grappling hook ready	Low			Internet			
Stake throw	Important	Kimmo	✓		✓		
Stake hit	Important	Aapo	✓		✓		
Stake pick up	Low			Internet			
Stake finisher	Important	Aapo	✓	Edit			
Heal	Important	Kimmo	✓	Internet			
Over heal	Low			Internet			
Player die	Important	Kimmo	✓	Internet			
Player take damage	Important	Kimmo	✓	Internet			
Player jump	Important	Kimmo	✓	Record		✓	
Player land	Medium	Kimmo	✓	Record			
Player fall into pit	Medium	Kimmo	✓	Internet			
Player dash	Important	Kimmo	✓	Record		✓	
Player slide	Important	Kimmo	✓	Record			
Player footsteps_1	Important	Aapo	✓		✓		
Player footsteps_2	Important	Aapo	✓		✓		
Player footsteps_3	Important	Aapo	✓		✓		
Player footsteps_4	Important	Aapo	✓		✓		
Enemy footsteps_1	Important	Aapo	✓	Internet			
Enemy footsteps_2	Important	Aapo	✓	Internet			
Enemy footsteps_3	Important	Aapo	✓	Internet			
Enemy footsteps_4	Important	Aapo	✓	Internet			
Enemy see player	Medium	Kimmo	✓	Record		✓	
Enemy die	Important	Aapo	✓		✓		
Enemy take damage	Important	Aapo	✓		✓		
Enemy projectile fly	Medium			Record			
Enemy projectile hit	Medium			Record			
Enemy Attack melee_1	Important	Kimmo	✓		✓		
Enemy Attack melee_2	Important	Kimmo	✓		✓		
Enemy Attack melee_3	Important	Kimmo	✓		✓		
Enemy shoot Ranged	Important			Record			
Enemy stun	Medium			Record			
Boss shout: Pirouette	Medium			Record			
Boss: Pirouette dagger fly				Internet			
Boss shout: blood servant's gr	Medium			Record			
Boss: blood servant effect audio				Internet			
Boss shout: Spike growth	Medium			Record			
Boss: Spike rise from floor				Internet			
Boss shout: Firewall	Medium			Record			
Boss: firewall effect				Internet			
Boss shout: Hellfire	Medium			Record			
Boss: hellfire effect				Internet			
Boss: vent activation (old cog with chains rotating)				Internet			
Door open/close	Low				✓		
Lever pull	Low			Internet			
Menu click	Low			Internet			